SECTION 3 | QUESTIONS WITH SOLUTIONS

Where Am I?

Maximum likelihood: score a guess, find the slope, climb

THE WEEK IN ONE LINE

Score a guess with the likelihood, ask for the slope, step uphill. This is the same climb in two settings: a fog game (find the top of an island) and a model (find where a phone is from Wi-Fi pings). The recipe is called *maximum likelihood*.

NOTATION USED BELOW

Symbol	Meaning
θ_1, θ_2	Parameters: the guess of where you stand, in meters east and north of $(0, 0)$. Together: θ.
D_i	Distance from the guess to tower i, in meters.
K_i	Pings caught from tower i, out of 100. Model: $K_i \sim \text{Binomial}(100, q(D_i))$.
$q(D)$	Chance that one ping is caught at distance D. It falls as D grows.
score	$\log_{10}$ of the likelihood: $\log_{10} \text{P}(\text{data} \mid \theta)$. Higher is better.
slope	$\partial\,\text{score}/\partial\theta_1$ and $\partial\,\text{score}/\partial\theta_2$: which way is uphill, and how steep.
step size	How far we walk per unit of slope. Here it is 6: new $\theta = \theta + 6 \times \text{slope}$.

QUESTION 1

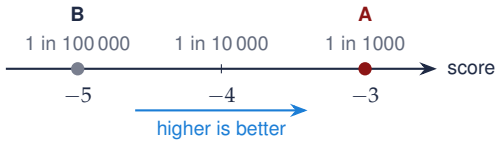
review

A model is a story with knobs. Story A has score -3 and story B has score -5. Which story is better?

SOLUTION

The score is $\log_{10}$ of the chance of the real data, so a score of s means the data had chance 10^s.

$$\text{A: } 10^{-3} = \tfrac{1}{1000}, \qquad \text{B: } 10^{-5} = \tfrac{1}{100\,000}, \qquad \frac{10^{-3}}{10^{-5}} = 10^2 = 100$$



ANSWER Story A (-3) is better. It makes the real data 100 times more likely. Every $+1$ on the score means 10 times more likely.

QUESTION 2 *fog game, round 1*

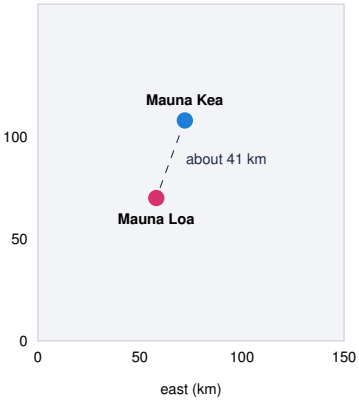
A real island is hidden under fog. The room has 7 readings, and each reading is the height at one spot. The room climbs and reaches a top. Is it the highest top?

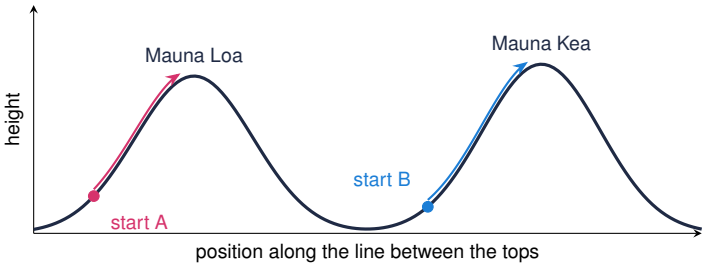
SOLUTION

Round 1 was Hawai'i Island, and it has **two tops**.

Top	Height	Spot (east, north) in km
Mauna Kea	4,207 m	(72, 108)
Mauna Loa	4,171 m	about (58, 70)

They are $4207 - 4171 = 36$ m apart in height and about 41 km apart on the map. A climb always ends on *a* top, but which top depends on where you start.





Start A climbs to Mauna Loa and start B climbs to Mauna Kea. Both climbs are correct, but only one finds the highest top.

After the climb, how could the room still find Mauna Kea?

Start somewhere else. Scout first. Or spend one reading far away (this is what the room's vote bought: scout far, or keep climbing here).

ANSWER Not necessarily. A climb finds *a* top, not always *the highest* top. Mauna Loa (4,171 m) is a perfectly good top, but Mauna Kea (4,207 m) is higher.

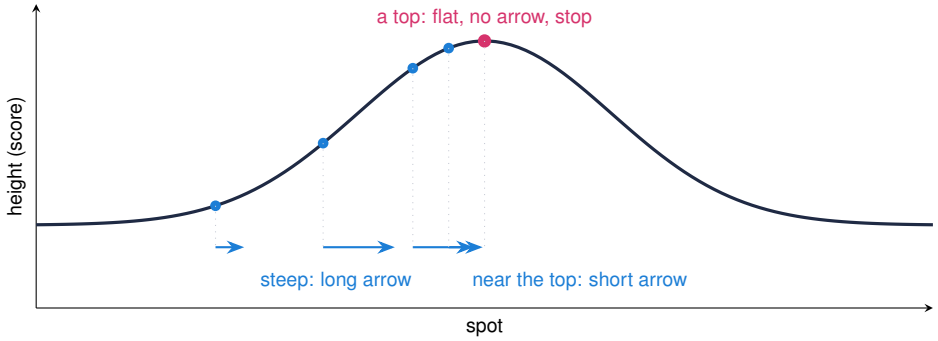
QUESTION 3 *fog game, round 2*

Round 2 is a new island, East Maui. Now each reading also points uphill, and the arrow is longer where the ground is steeper. What did the arrows change?

SOLUTION

Round 2 had one top: Haleakalā, 3,051 m at $(24, 19)$.

An arrow is the **slope**: it points the way uphill. With arrows you know which way to step, so you no longer guess. The arrows also get shorter as the ground flattens, and at a top there is no uphill left.



ANSWER The arrows give the **direction** (which way is uphill) and the **size** (how steep). They shrink toward the top, and at the top they vanish: flat in every direction means stop.

QUESTION 4 *key ideas*

Name each idea from the fog game in the language of machine learning.

SOLUTION

In the fog game	In machine learning
A spot (east, north)	the **parameters** θ
Its height	the **score**
The way uphill	the **slope**
How far you walk	the **step size**
Flat: every nudge is lower	a top: **stop**
Mauna Loa	is this the **highest** top?

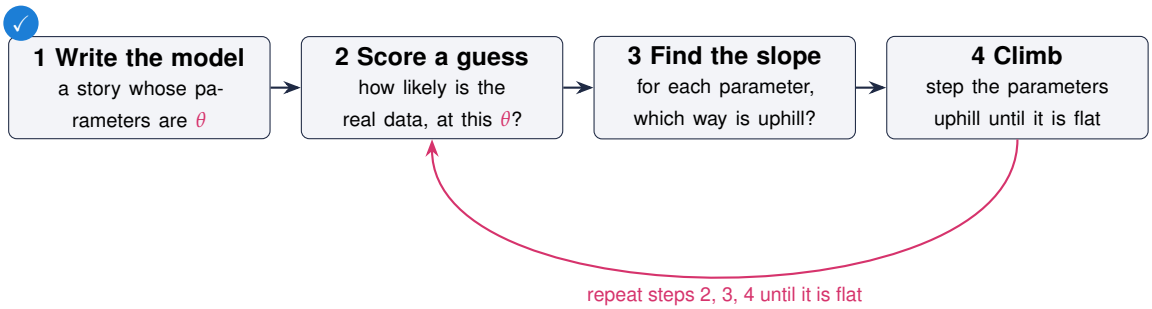
ANSWER The game is exactly how a computer learns: choose parameters, read the score, follow the slope, and stop on a top.

QUESTION 5 *think*

Your phone catches Wi-Fi pings from three towers. What are the main steps to find the phone?

SOLUTION

This is the recipe the room should reach. It is called **maximum likelihood** (MLE).



Step 1 is already done once the model is written. The rest of the section is steps 2, 3 and 4, one at a time.

ANSWER Write the model, score a guess, find the slope, climb. In one line: score a guess with the likelihood, ask for the slope, step uphill.

QUESTION 6 *step 1: the model*

Write the model for "Where am I?". Three towers stand on an 80×50 m lawn at known spots. Each tower sends 100 pings in ten seconds. What are the parameters, and how are the pings produced?

SOLUTION

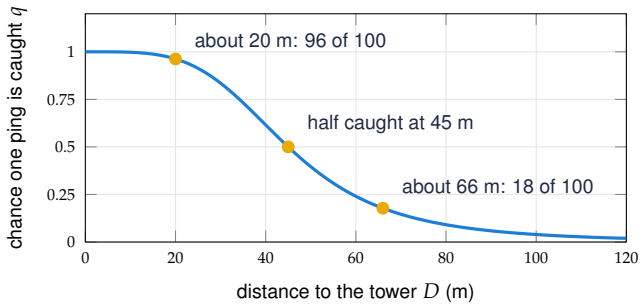
The story. (1) Pick a spot where you might be. (2) Each tower is some distance from that spot. (3) The farther the tower, the smaller the chance each ping gets through (radio signals weaken with distance). (4) Each tower sends 100 pings, each one caught or missed with that chance, so the number caught is binomial.

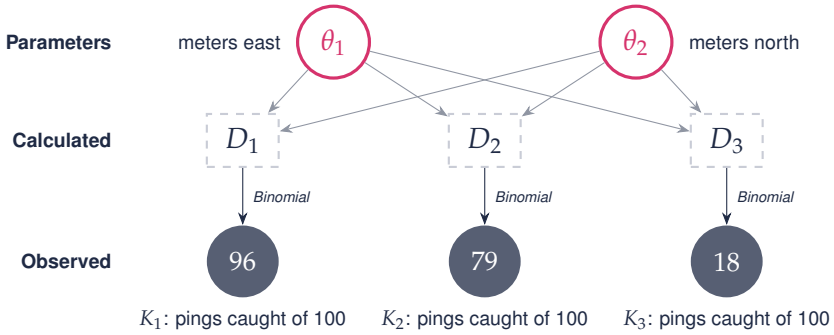
Parameters (what we do not know): θ_1 meters east and θ_2 meters north.

$$D_i = \sqrt{(\theta_1 - x_i)^2 + (\theta_2 - y_i)^2}, \qquad K_i \sim \text{Binomial}\big(100,\ q(D_i)\big)$$

Tower i	Stands at $(x_i,\ y_i)$	Pings caught K_i (of 100)
1	(38, 46)	96
2	(74, 10)	79
3	(6, 6)	18







Note. The slides show the catch curve only as a picture. The curve drawn here is $q(D) = 1/\left(1 + (D/45)^4\right)$. It is my fit, and it reproduces every number on the slides (for example $q(41.2) = 0.59$ and about 64 expected pings from tower 3 at $(40, 25)$). The course may write its own curve differently.

ANSWER Parameters: θ_1 (east) and θ_2 (north). For each tower, the distance D_i sets the chance $q(D_i)$, and the pings caught are $K_i \sim \text{Binomial}(100, q(D_i))$.

QUESTION 7 *step 2: score a guess*

Take the guess $\theta = (70, 20)$. Tower 1 stands at $(38, 46)$ and the phone really caught 96 of its 100 pings. If you stood at $(70, 20)$, how likely are the real pings from tower 1? Give the score.

SOLUTION

Step 1: distance and catch chance

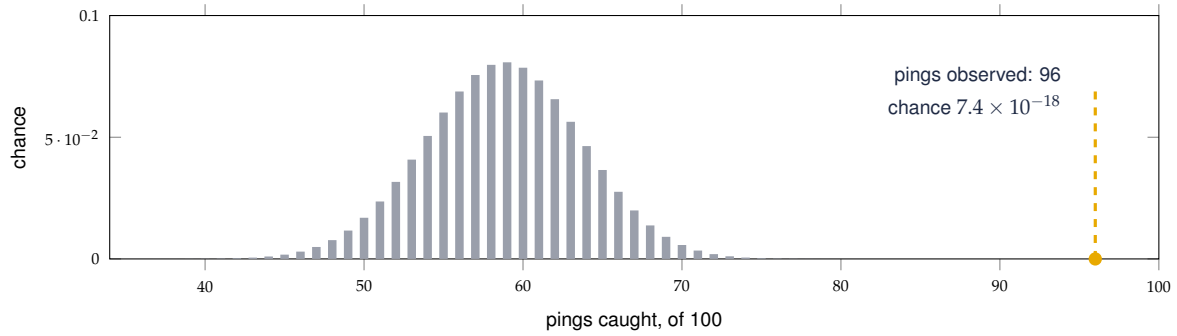
$$D_1 = \sqrt{(70-38)^2 + (20-46)^2} = \sqrt{1700} \approx 41.2 \text{ m} \quad \Longrightarrow \quad q \approx 0.59$$

So the pings caught would follow Binomial$(100, 0.59)$. Most of the time you would catch about 59.

Step 2: the likelihood is the chance of exactly 96

$$\mathrm{P}(K_1 = 96 \mid \theta) = \binom{100}{96} q^{96}(1-q)^4 = 3\,921\,225 \cdot q^{96}(1-q)^4 \approx 7.4 \times 10^{-18}$$

This chance is the **likelihood**, and it is our score for the guess. It is tiny: the data is telling us "probably not here".



Step 3: take the log

$$\text{score} = \log_{10}(7.4 \times 10^{-18}) = -17.13$$

Scores are negative because a chance is below 1. Higher is still better: -3 beats -17.

Rounding note. The slide's 7.4×10^{-18} uses the unrounded catch chance 0.5875 (distance 41.2 m), which is shown as 0.59. If a student types Binomial$(100, 0.59)$ into a calculator, they get 1.1×10^{-17} (score -16.95). With the exact distance 41.23 m the score is -17.18. A far-tail chance moves a lot when the catch chance moves a little, so quote scores to about one decimal.

ANSWER Likelihood $\approx 7.4 \times 10^{-18}$, score $= -17.13$. Why take the log at all? The chances are absurdly small, and the log keeps numbers readable while the best guess stays in the same place.

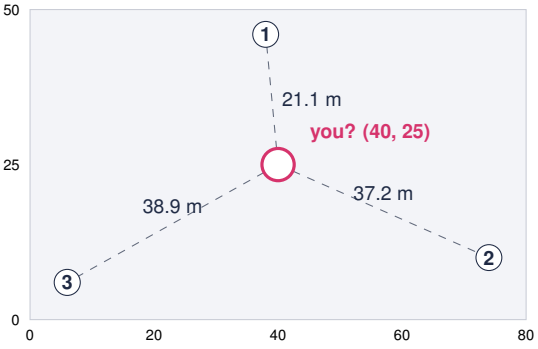
QUESTION 8 *three towers*

The phone also caught 79 of tower 2's pings and 18 of tower 3's. Score the guess $\theta = (40, 25)$ using all three towers.

SOLUTION

The towers ping on their own, so the chances of the three records **multiply**, and the score **adds the logs**.

$$\text{likelihood} = \prod_{i=1}^{3} \mathrm{P}(K_i = k_i \mid \theta), \qquad \text{score} = \sum_{i=1}^{3} \log_{10} \mathrm{P}(K_i = k_i \mid \theta)$$



Expected pings are $100\,q(D_i)$. Tower 1 caught about what we expect. Tower 2 caught more than expected. Tower 3 caught *far fewer* than expected, which already says "you are not this close to tower 3".

Tower	at	D_i (m)	q	expected	caught k_i	$\mathrm{P}(K_i = k_i)$	$\log_{10}$
1	$(38, 46)$	21.1	0.954	95.4	96	0.191	-0.719
2	$(74, 10)$	37.2	0.683	68.3	79	0.00552	-2.258
3	$(6, 6)$	38.9	0.641	64.1	18	3.70×10^{-21}	-20.431

$$\text{score} = -0.719 - 2.258 - 20.431 = -23.41$$

$$\text{likelihood} = 0.191 \times 0.00552 \times 3.70 \times 10^{-21} \approx 3.9 \times 10^{-24}$$

Rounding note. Slide 39 shows -23.53 at the same spot $(40, 25)$ because it multiplies numbers rounded from rounded distances (21.1, 37.2, 38.9 m). The exact score is -23.41, and it is the number the climb starts from (slide 44). The 3.7×10^{-21} for tower 3 becomes 2.9×10^{-21} after that rounding.

ANSWER At $(40, 25)$ the score is ≈ -23.41. A good guess is one that makes *all three* real counts likely at once.

QUESTION 9 *step 3: find the slope*

The Code Agent gives the slope of the score. Read the two numbers at $\theta = (40, 25)$: which way is uphill, and how steep?

SOLUTION

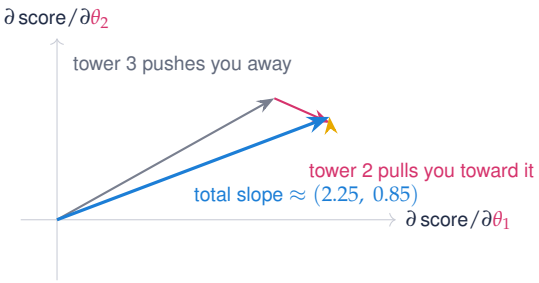
Nobody derives the slope in section. The computer writes it:

$$\frac{\partial\,\text{score}}{\partial\theta_1} = 1.74 \sum_{\text{towers}} \frac{(\text{expected} - \text{caught})(\theta_1 - \text{tower east})}{\text{distance}^2}$$

$$\frac{\partial\,\text{score}}{\partial\theta_2} = 1.74 \sum_{\text{towers}} \frac{(\text{expected} - \text{caught})(\theta_2 - \text{tower north})}{\text{distance}^2}$$

Each tower contributes one term. Here is every term at $(40, 25)$:

Tower	$(\theta_1 - x_i,\ \theta_2 - y_i)$	D_i^2	expected	caught	exp. − caught	θ_1 term	θ_2 term
1	$(+2, -21)$	445	95.4	96	-0.6	-0.005	$+0.050$
2	$(-34, +15)$	1381	68.3	79	-10.7	$+0.460$	-0.203
3	$(+34, +19)$	1517	64.1	18	$+46.1$	$+1.796$	$+1.004$
					sum	**2.252**	**0.850**



Tower 3 caught only 18 but we would expect about 64, so we must be too close to it. It pushes us away, toward the east and north. Tower 2 caught more than expected, so it pulls us toward it. Tower 1 is almost silent. A tower that caught more than expected pulls; one that caught fewer pushes.

ANSWER $\frac{\partial\,\text{score}}{\partial\theta_1} \approx 2.25$ and $\frac{\partial\,\text{score}}{\partial\theta_2} \approx 0.85$. Both are positive, so both east and north are uphill, and **east is steeper**.

QUESTION 10 *step 4: climb*

Use step size 6. Starting at $\theta = (40, 25)$, hop uphill: new $\theta = \theta + 6 \times$ slope. Where does the climb stop, and what is the score there?

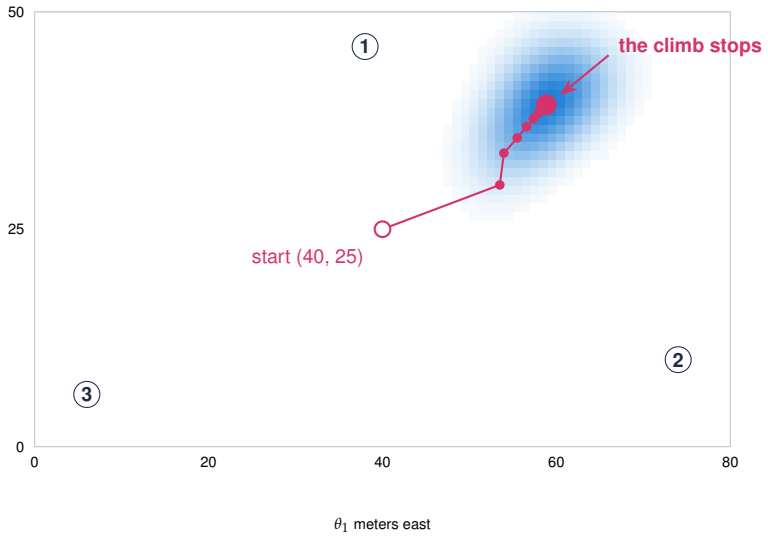
SOLUTION

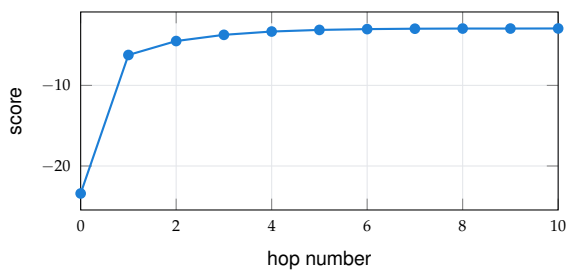
The first hop, out loud

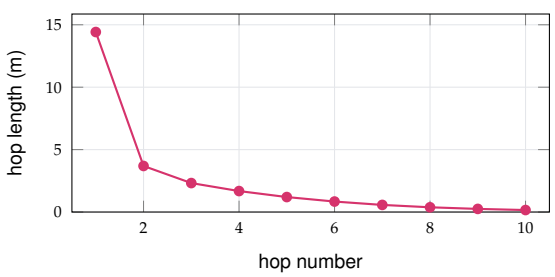
$$6 \times 2.25 = 13.5 \text{ m east}, \qquad 6 \times 0.85 = 5.1 \text{ m north} \quad \Longrightarrow \quad \theta = (53.5,\ 30.1)$$

The score jumps from -23.41 to -6.23. Then we ask for the slope again and hop again. The full climb:

Hop	slope E	slope N	hop E (m)	hop N (m)	new east	new north	score
0					40.00	25.00	−23.41
1	2.248	0.849	13.49	5.09	53.49	30.09	−6.23
2	0.079	0.609	0.47	3.65	53.96	33.75	−4.50
3	0.256	0.291	1.53	1.75	55.49	35.49	−3.73
4	0.179	0.216	1.07	1.30	56.57	36.79	−3.32
5	0.132	0.151	0.79	0.91	57.36	37.70	−3.12
6	0.093	0.104	0.56	0.62	57.92	38.32	−3.02
7	0.064	0.070	0.38	0.42	58.30	38.74	−2.97
8	0.043	0.046	0.26	0.28	58.56	39.01	−2.95
9	0.028	0.030	0.17	0.18	58.73	39.19	−2.95
10	0.018	0.019	0.11	0.12	58.84	39.31	−2.94







At the top, every nudge of 1 m in any direction reads lower, and both slopes are about 0.01: **a top, stop**. After 10 hops the next hop would be shorter than 10 cm.

Note. The table is recomputed from the model with exact slopes, so the final point reads $(58.84, 39.31)$. The live slide shows $(58.84, 39.34)$. The 0.03 m gap is rounding and does not change anything.

ANSWER The climb stops at $\approx (58.8,\ 39.3)$ with score ≈ -2.94 after 10 hops.

QUESTION 11 *the reveal*

The phone was really at $(60, 38)$. How far apart are the climb and the phone? Why did the climb not land exactly on the phone?

SOLUTION

$$\sqrt{(60 - 58.84)^2 + (38 - 39.31)^2} \approx 1.8 \text{ m}$$

The pings are random, and three counts cannot pin the phone down to the centimeter. Compare what each spot *expects* with what the phone *caught*:

	tower 1	tower 2	tower 3	score
caught (real data)	96	79	18	
expected at the phone $(60, 38)$	93	81	21	-3.18
expected at the climb's top	95	78	21	-2.94

The top is the spot whose expected counts match the real counts best, so its score (-2.94) is higher than the score at the true spot (-3.18). Another 300 pings from the same phone would move the top somewhere else nearby.

ANSWER 1.8 m apart. Randomness in the pings: from 96, 79 and 18 the most likely spot is about $(59, 39.5)$, and the phone sat 1.8 m from where the climb stopped.

QUESTION 12 *exit ticket*

One word each. (1) Higher score: better or worse? (2) Tower 1 caught 96 of 100. Is your phone near it or far? (3) Every nudge reads lower. Hop or stop?

SOLUTION

	Answer	Why
1	**Better**	The score is how likely the real pings are from your guess.
2	**Near**	Close to a tower, almost all of its pings arrive.
3	**Stop**	You are on a top.

ANSWER Better, near, stop. The week in one line: score a guess with the likelihood, ask for the slope, step uphill.

COMMON QUESTIONS AND MISTAKES TO WATCH FOR

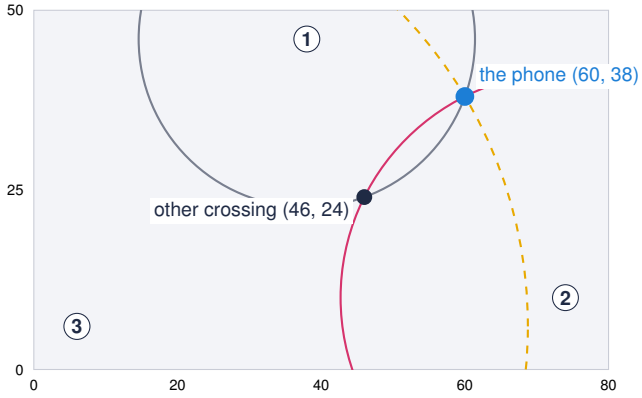
Question or mistake	Answer
Why is the score negative?	It is the log of a chance, and a chance is below 1. Closer to 0 is better, and $+1$ means 10 times more likely.
Which is better, -3 or -17?	The higher one, -3.
Why multiply the towers?	Each tower pings on its own, so the chances of the three records multiply. Logs turn that product into a sum.
Why take the log at all?	The chances are absurdly small (7.4×10^{-18}). The log keeps the numbers readable, and the best guess is in the same place either way.
Why did the hops shrink if the step size never changed?	A hop is step size $\times$ slope, and the slope shrinks near the top.
Hopping by the slope itself.	A slope of 0.85 means a hop of $6 \times 0.85 = 5.1$ m, not 0.85 m.
Taking a top for the highest top.	Mauna Loa is a perfectly good top. It is just not Mauna Kea.
Why did the climb not land exactly on the phone?	The pings are random, and three counts cannot pin it down exactly.

QUESTION 13 *if you have extra time*

(a) Why does a phone listen to many towers, not one? (b) What would a step size of 60 do?

SOLUTION

(a) With one tower, the score depends only on the distance to it. Every spot at the same distance scores exactly the same: a whole *ring* of equally good guesses. Two towers leave two spots where their rings cross. The third tower picks one.



Rings of tower 1 (solid grey) and tower 2 (pink) cross at $(60, 38)$ and $(46, 24)$. Tower 3's dashed ring passes through $(60, 38)$ only. This is the ideal, noise-free picture. Real pings are random, so each ring is a blurry band.

(b) The first hop would be $60 \times (2.25,\ 0.85) \approx (135,\ 51)$ m, from $(40, 25)$ to about $(175,\ 76)$, far off an 80×50 m lawn. A step that is too big jumps past the top and bounces around it. A step that is too small takes forever.

ANSWER (a) One tower gives only a ring of equally good guesses. Three towers pin down one spot. (b) A step size of 60 overshoots the top.